

Mortgage pools: 505 priced — and a possible way out of buying deal data

2026-09-30 · valuation date 2009-03-31 · control date 2009-06-10

Plain language throughout; every mortgage term is defined where it first appears. There is a short appendix at the end with the commands, which you can ignore. Azure is **not** in this report — it has a separate one-page note, because it is a different question.

1. The short version

The mortgage data you sent came back complete, and we built the pricer around it. **505 of the 882 mortgage securities are now priced**, each with a spread, a duration and an average life.

But the more important result is about the **other 377**, and it is not what we expected in July.

A third of the whole portfolio — 756 securities — are slices of structured deals, and a slice cannot be priced from its own description. Until this week the only route we could see was buying the deal models. There may be a cheaper one, and the request to test it is already with Liping.

That is the decision in this report. Everything else is how we got there.

2. What we can price now, and what the numbers say

A **mortgage pass-through pool** is the simple case: a few thousand household mortgages, bundled, with every dollar of interest and principal passed straight through to the holder in proportion. No priority, no slices. If you know the pool's rate, its remaining term and how fast borrowers repay, you know the cash flows.

490 of the 882 are pools like that, and 478 of them price. A further **27** are **TBA**s — a forward contract to receive a pool that has not been delivered yet, quoted by coupon and settlement month rather than by a named pool — which we price as forwards. **505 in total.**

2.1 The one number we cannot observe, and what we did about it

Everything about a mortgage turns on **prepayment**: the borrower's right to repay early, usually by refinancing. It is measured as **CPR** — the percentage of the remaining balance that repays ahead of schedule in a year.

⚠ We do not have the 2009 prepayment speeds. The Bloomberg data was pulled this year, so its speeds describe today's housing market rather than March 2009's. Rather than pick a number and present one answer as if we knew it, **we priced every pool at three speeds and report the trade-off:**

if prepayment is...	median spread	slowest 10%	fastest 10%
15% a year	265 bp	182	326
25% a year	250 bp	170	298
35% a year	220 bp	148	267

From `outputs/pool_risk_2009-03-31.csv`, all 505.

Read it as: the spread is somewhere around 220–265 bp, and the prepayment assumption moves it by roughly 25 bp for every 10 points of speed. That band is the honest width of what we know today. Liping's data collapses it to a single column.

⚠ **One number is deliberately in the output and deliberately not used.** If we assume the credit spread is zero — arguable, since these carry a government guarantee — then the prepayment speed that would explain the observed prices comes out at a **median 66% a year**, and for 48 securities no speed explains the price at all. **66% is not a real prepayment rate.** It is the arithmetic telling us the market was demanding a genuine spread even on guaranteed paper in March 2009, which is what one would expect that month. We keep it as evidence, not as an input.

2.2 The model reproduces something nobody told it

How far a pool's own mortgage rate sits above what a borrower could refinance into today is the single best predictor of whether they will refinance. These pools sit a **median 1.53 percentage points above** the prevailing rate.

Sorting all 505 by that gap and taking the median spread in each band:

pool rate above market rate by...	securities	median spread
less than 0.5pp	66	173 bp
0.5 to 1.5pp	176	217 bp
1.5 to 2.5pp	190	274 bp
more than 2.5pp	73	296 bp

★ **The spread rises steadily with the refinancing incentive, across all 505, and nothing in the model was told to make that happen.** This is the shape mortgage spreads are known to have — the market charges more for the bonds whose borrowers are most likely to leave early. Getting it out of a model that only knows cash flows and a discount curve is the strongest single sign that the pricing is behaving sensibly.

2.3 A second check, where the answer flips the other way — correctly

Almost all these pools trade **above face value** (median price 104.63): they pay more than today's mortgage rates, so they are worth more than the balance outstanding. **Nine trade below.**

Now, this next point reads backwards at first, so it is worth a slow sentence. We are solving for the spread that makes our valuation *equal the price you are actually marked at*. For a pool trading above face value,

faster prepayment is bad news — the above-market interest stops sooner — so our valuation falls. But the price we have to match has not moved. **So the spread has to come down to lift the valuation back up.** For a pool trading below face value, everything reverses.

	securities	spread moves per +10 points of prepayment
priced above face value	496	–26 bp
priced below face value	9	+32 bp

★ **Both directions are right, and the split is 496 against 9** — those nine are the entire counter-example in the book, and the model finds them.

2.4 Risk numbers, and one honest disagreement

Median **average life 3.25 years**. Median **spread duration 2.80 years** — how much the price moves when the spread moves.

⚠️ Against the custodian's own duration figure, the median gap is **0.76 years, and 102 of 478 differ by more than 1.5 years**. That is less a reconciliation failure than two different quantities. Ours holds the prepayment speed fixed; theirs appears to let prepayment respond to rates. We publish both columns side by side rather than choose. **It is one more reason the 2009 speeds matter.**

3. The 377 we do not price, each one named

Nothing is quietly dropped. Every one of the 882 appears either in the priced output or in a companion file with a reason beside it.

	count	why
priced	505	pools and TBAs
slices of structured deals	192	see section 4
interest-only strips	76	receives the interest and no principal at all
principal-only strips	76	receives the principal and no interest at all
description matches nothing we recognise	18	we cannot tell what it is
adjustable-rate pools	13	the rate resets, and the rate we hold is this year's
TBAs that do not state a settlement month	2	a gap in the description text

505 + 344 + 33 = 882.

⚠️ **The 18 we cannot recognise are a real gap, not a rounding line.** We read the custodian's description text; for these it matches nothing we know, so we decline rather than guess. Guessing would produce a confident price for a security whose structure we had not actually identified — the worst of the available outcomes.

4. The third of the book that needs deal data

Interest-only strips, principal-only strips, REMICs and CMOs are all the same idea: one pool's cash flows are cut into pieces and handed to different investors under a set of priority rules written for that one deal. Which piece gets paid, in what order, under what conditions — all of it is specific to that deal, and none of it is in any data feed we can subscribe to field by field.

Counting that same structure across every securitised class:

class	securities	slices of a deal
Government MBS	882	344
Non-Government CMOs	264	264
Commercial Mortgage-Backed	69	69
Asset-Backed Securities	79	79
total		756 — a third of the 2,260-security portfolio

⚠ **And they are not 756 similar problems.** In Government MBS alone, the 344 slices come from **170 different deals** — and **144 of those deals contribute exactly one slice to your portfolio**. To work out what one slice receives, you need the rules for the *whole* deal, including every other slice you do not own. So the requirement is roughly **170 complete deal models**, not 344 security records. That is the shape of the purchase.

⚠ **The obvious shortcuts do not work, and we checked two rather than assume.** The letter in a slice's name looks like it should classify it, but the lettering is each deal's own convention: of the 54 whose letter begins with **P**, **29 turn out to be principal-only strips** rather than the scheduled-payment slices the letter suggests. And the custodian's own duration on these ranges from **–102 to +57 years**, against a well-behaved 1 to 4 on the ordinary pools. Those extremes are *correct* — an interest-only strip really does gain value when rates rise, because the borrowers stop leaving — which is exactly why treating these as ordinary pools would be a category error rather than a rough approximation.

4.1 ★ The possible way out

Since July the plan has been that this needs a deal-structure purchase. **We now think we were asking for the wrong thing.**

The only job those deal rules do is decide **how much principal each slice receives each month**. And there is a published monthly number that records exactly that decision after the fact: the **factor** — the share of a security's original balance still outstanding.

From one month's factor and the next:

- **principal paid** = original size × (last month's factor – this month's factor)
- **interest paid** = original size × last month's factor × rate ÷ 12

★ **That is the complete cash flow, and it needs no knowledge of the deal's rules at all.** It also handles every structure without special treatment: a principal-only strip has no interest term, an interest-

only strip has no principal term, and a slice that rolls its interest into its own balance instead of paying it works out to exactly zero cash that month — which is precisely what such a slice does.

☆☆ **And because we are valuing as at 2009, most of that history has already happened.** Of the 344 slices: **89 have paid off completely**, so their cash flow is a matter of record with no assumption of any kind; **251 are still outstanding**, with a median 6.7 years left against the 17 years we can already observe. We already hold the starting factor for **341 of the 344**.

This is the cheaper request, not merely a different one. A factor is a published monthly fact with a date on it — ordinary historical data. A *forecast* of future cash flows, by contrast, is computed from today's loan pool and cannot be run backwards to 2009.

⚠ **The honest limit, and it must travel with every number this produces.** Using what actually happened is hindsight: it gives the return the holder really earned, not the spread the market was demanding in March 2009. For **risk** measures — duration, sensitivity — a cash flow schedule is a cash flow schedule, and the numbers stand on the same footing as the pools' figures above. For "*what was this worth to the market that morning*", the deal structure is still required.

☆☆ **The test is built into the request.** It includes **20 ordinary pools we already price**, so the method gets checked against numbers we already trust *before* it is pointed at a single slice.

4.2 What this does to the purchase decision

It does not answer the question; it adds a branch that costs almost nothing to test. The request is with Liping — 376 securities, one lookup each. If the factor history comes back, we will know within days whether these 756 securities can be valued without buying deal models. If it does not, the purchase question returns exactly as it stood, and we will have spent a few hours finding out.

5. Three modelling choices we made, and why

We solve for the prepayment speed rather than the spread. One observed price can identify one unknown, not two. For government-guaranteed paper the credit spread is defensibly near zero, so spending that single equation on prepayment is the better use of it — and when the 2009 speeds arrive, the two become a cross-check on each other instead of alternatives.

We refuse to report a prepayment speed when it carries no information. For some pools the price barely moves across the entire range of possible speeds — less than the precision the price is even quoted to. The arithmetic still produces an answer, and that answer is determined by rounding rather than by prepayment. We decline to print it.

A TBA settles on a stated future month, and the description does not give the year. A contract that says "settles April" is a one-month forward at the March valuation date and an *already-delivered* position at the June control date — not a fourteen-month forward. Running both dates is what surfaced that, and it is why we run both.

6. How we know nothing else moved

This round added a mortgage pricer and touched nothing else. To demonstrate that rather than assert it, all four existing valuation runs were repeated and compared: **not one number changed in any of them** — the corporate book, the government and sovereign book, the agency and inflation-linked book, or the callable book. **656 automated checks pass.**

Appendix — running it

```
PYTHONPATH=src python scripts/pool_risk.py          # 2009-03-31
FIP_VAL_DATE=2009-06-10 PYTHONPATH=src python scripts/pool_risk.py
```

Produces `outputs/pool_risk_<date>.csv` and a companion `pool_disposition_<date>.csv` with one row per security in the class, so the two files together always account for all 882.

The mortgage engine sits in the same structure as the rest of the library — the mathematics in one layer, the per-product functions with their inputs listed one per line in another — the arrangement agreed in August and used for the corporate and government books.